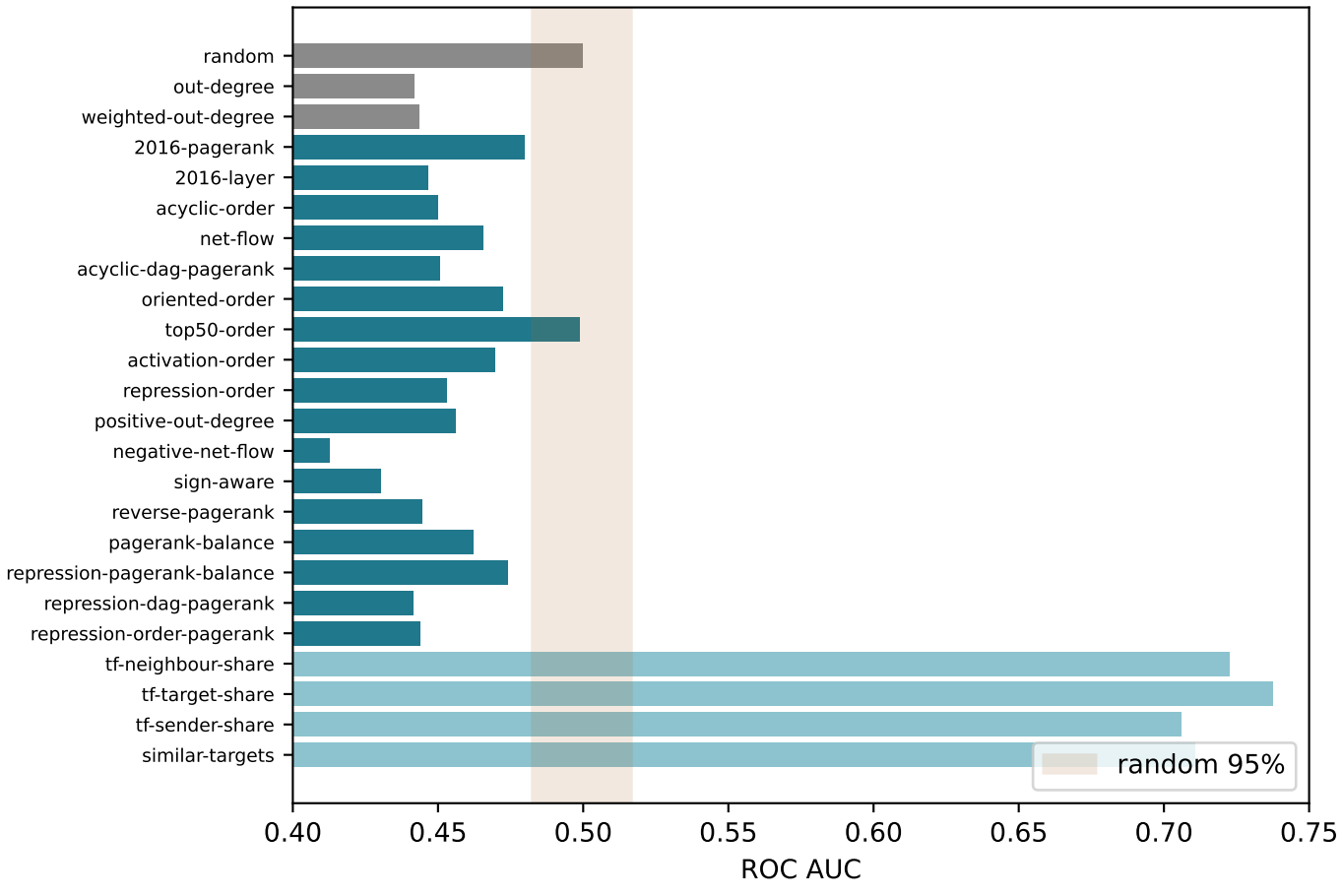
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,980 genes ranked, 1,155 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	9.6 (4–16)	0.500 (0.482–0.517)				
out-degree	1	0.442	1.000	1.000	-0.002 (-0.003, -0.001)	0.002
weighted-out-degree	2	0.444	1.000	1.000	+0.000 (+0.000, +0.000)	
2016-pagerank	10	0.480	0.494	0.987	+0.036 (+0.020, +0.052)	< 0.001
2016-layer	7	0.447	0.861	1.000	+0.003 (-0.014, +0.020)	0.690
acyclic-order	1	0.450	1.000	1.000	+0.007 (-0.001, +0.015)	0.104
net-flow	1	0.465	1.000	1.000	+0.022 (+0.011, +0.033)	< 0.001
acyclic-dag-pagerank	3	0.451	0.997	1.000	+0.007 (+0.001, +0.013)	0.014
oriented-order	1	0.472	1.000	0.999	+0.029 (+0.016, +0.043)	< 0.001
top50-order	12	0.499	0.244	0.552	+0.055 (+0.037, +0.073)	< 0.001
activation-order	10	0.469	0.494	1.000	+0.026 (+0.016, +0.035)	< 0.001
repression-order	4	0.453	0.991	1.000	+0.009 (-0.004, +0.024)	0.194
positive-out-degree †	8	0.456	0.763	1.000	+0.012 (+0.009, +0.016)	< 0.001
negative-net-flow †	2	0.413	1.000	1.000	-0.031 (-0.044, -0.017)	< 0.001
sign-aware †	2	0.430	1.000	1.000	-0.013 (-0.020, -0.006)	< 0.001
reverse-pagerank †	1	0.444	1.000	1.000	+0.001 (-0.002, +0.004)	0.522
pagerank-balance †	3	0.462	0.997	1.000	+0.019 (+0.005, +0.033)	0.004
repression-pagerank-balance †	11	0.474	0.359	0.998	+0.030 (+0.017, +0.045)	< 0.001
repression-dag-pagerank †	9	0.441	0.633	1.000	-0.002 (-0.016, +0.011)	0.792
repression-order-pagerank †	1	0.444	1.000	1.000	+0.000 (-0.012, +0.013)	0.936
tf-neighbour-share ‡	100	0.723	< 0.001	< 0.001	+0.279 (+0.256, +0.302)	< 0.001
tf-target-share ‡	96	0.737	< 0.001	< 0.001	+0.294 (+0.269, +0.318)	< 0.001
tf-sender-share ‡	100	0.706	< 0.001	< 0.001	+0.262 (+0.239, +0.286)	< 0.001
similar-targets ‡	97	0.711	< 0.001	< 0.001	+0.267 (+0.240, +0.290)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

collectri

Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	31,889	0.042 (2.89×)	3.59×	1.07×	0.79 (0.78)
resolve_reciprocals → drop_sinks → prune_triangles	30,991	0.005 (1.22×)	2.23×	1.10×	0.34 (0.33)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	22,885	0.004 (1.36×)	2.42×	1.32×	0.28 (0.29)
absolute	31,889	0.042 (2.89×)			
absolute → minimum_feedback_order	31,040	0.030 (2.56×)			
orient_reciprocals ratio=2.0 → absolute	30,992	0.018 (1.92×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	30,992	0.017 (1.90×)			
top_outgoing k=50 → absolute	31,889	0.015 (4.26×)			
top_outgoing k=50 → absolute → minimum_feedback_order	31,010	0.010 (3.67×)			
keep_sign sign=1 → absolute	31,889	0.038 (3.59×)			
keep_sign sign=1 → absolute → minimum_feedback_order	31,040	0.026 (3.13×)			
keep_sign sign=-1 → absolute	29,978	0.005 (1.06×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	29,647	0.004 (1.10×)			
keep_sign sign=-1	29,978	0.005 (1.06×)		1.06×	0.21 (0.21)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 2,445,002 edges (832,058 negative), largest strong component 11,980.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 2,445,002 edges (832,058 negative), largest strong component 11,980.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 827,854 edges (768,518 negative), largest strong component 11,766.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 827,854 edges (768,518 negative), largest strong component 11,766; DAG keeps 572,320 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,445,002 edges (0 negative), largest strong component 11,980; DAG keeps 1,947,664 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 2,445,002 edges (0 negative), largest strong component 11,980.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 2,445,002 edges (0 negative), largest strong component 11,980; DAG keeps 1,947,664 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,570,776 edges (0 negative), largest strong component 11,494; DAG keeps 1,541,310 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 522,835 edges (0 negative), largest strong component 11,971; DAG keeps 384,056 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,612,944 edges (0 negative), largest strong component 11,968; DAG keeps 1,198,025 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 832,058 edges (0 negative), largest strong component 10,987; DAG keeps 758,793 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 2,445,002 edges (832,058 negative), largest strong component 11,980.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 2,445,002 edges (832,058 negative), largest strong component 11,980.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 2,445,002 edges (832,058 negative), largest strong component 11,980.

reverse-pagerank: Highest reverse PageRank (raw graph)